europass

Houdaifa El Boukhari

ID: KB218600 | **Date of birth:** 28/11/2003 | **Place of birth:** tangier, Morocco | **Nationality:** Moroccan |

Phone number: (+212) 607932833 (Mobile) | **Email address:** houdaifaelboukhari1@gmail.com | **LinkedIn:** Houdaifa El Boukhari | **Github:** houdaifa-boukhari |

Address: hay el berouaka rue 20 No 5, 90090, tangier, Morocco (Home)

ABOUT ME

I am a passionate and self-motivated IT student specializing in Information Systems and Network Architecture. Currently studying at **1337 School**, part of the global **42 Network**, I've built strong technical and interpersonal skills through an intensive, peer-driven learning environment.
My hands-on experience spans **C programming**, **C++**, **Unix/Linux systems**, and **network architecture fundamentals**. I actively engage in **Capture The Flag (CTF)** cybersecurity competitions, where I focus on **reverse engineering**, **binary exploitation**, and **cryptography** — sharpening both my technical depth and teamwork capabilities through real-world challenges.
I am particularly interested in **low-level programming**, **system internals**, and **offensive security**, and I'm eager to apply these skills in a professional setting. I'm currently seeking **entry-level opportunities or internships** where I can grow as a **cybersecurity analyst**, **vulnerability researcher**, or **low-level developer**, contributing to impactful projects and learning from experienced teams.

EDUCATION AND TRAINING

04/12/2023 – CURRENT Tetouan, Morocco

RNCP LEVEL 7 – EXPERT IN IT ARCHITECTURE – INFORMATION SYSTEMS AND NETWORK ARCHITECTURE

1337 School – Member of 42 Network

Website https://1337.ma/ | **Level in EQF** EQF level 7

LANGUAGE SKILLS

Arabic – Native Language

Full professional fluency in speaking, reading, and writing.

English – Intermediate (B1)

Comfortable with reading and listening in technical contexts. Actively improving spoken and written fluency through collaborative coding projects, CTF events, and documentation work.

SKILLS

Technical / Digital Skills

42 Common Core (Problem-solving, Algorithms, Shell, Git, C programming) | CTF challenges | Linux / Unix Systems | Python / Bash scripting | CYBER SECURITY | C++ (OOP, STL)

Soft Skills

Teamwork | Problem Solving | Adaptability | Time Management | Critical Thinking | Communication | Public Speaking | Self-discipline

PROJECTS

CURRENT

Inception (Docker-Based System Administration)

- Architected and deployed a Docker Compose–driven infrastructure on a VM, including:
    - **NGINX** reverse-proxy with TLS 1.2/1.3 as sole entry point

- **WordPress** container with PHP-FPM
- **MariaDB** container with persistent volumes for both database and website files
- Custom docker-network for secure inter-container communication

- Implemented automated container restarts, environment-variable management via .env files and Docker Secrets, and a Makefile to orchestrate builds—culminating in a production-grade, self-contained stack.

Link https://github.com/houdaifa-boukhari/Inception

C++ Modules 00–09 – Advanced Object-Oriented Programming

Solo Projects | Language: C++98 | **Domain:** Object-Oriented Programming, Memory Management, Templates, STL
Completed a series of progressively challenging modules designed to deepen understanding of C++ programming concepts and best practices.

- **Modules 00–01:** Introduced to fundamental concepts such as namespaces, classes, member functions, standard I/O streams, initialization lists, static and const qualifiers, memory allocation (stack vs. heap), pointers, references, and file streams.
- **Modules 02–04:** Explored ad-hoc polymorphism, operator overloading, canonical class form, inheritance, subtype polymorphism, abstract classes, and interfaces.
- **Modules 05–06:** Focused on exception handling, repetition structures, and the implementation of custom exception classes to manage error handling effectively.
- **Modules 07–09:** Delved into advanced topics including templates, STL containers (such as vector, map, and set), iterators, and algorithms. Applied these concepts to practical projects like a Bitcoin exchange rate parser and a Reverse Polish Notation calculator.

- Ensured code quality and resource safety by strictly following the Orthodox Canonical Form and best practices.

Link https://github.com/houdaifa-boukhari/CPP_42-School

Cub3D – 3D Raycasting Game Engine

Team Project | Language: C | **Technologies:** MiniLibX, Math, Graphics Programming
Created a 3D maze exploration game using raycasting principles, inspired by Wolfenstein 3D.

- Built a custom rendering engine to simulate first-person perspective inside a 2D map.
- Parsed .cub configuration files and implemented dynamic navigation (WASD + arrow keys).
- Added advanced features: wall collision, animated sprites, doors, and a real-time minimap.

Engineered an interactive graphical environment demonstrating strong understanding of mathematics, memory, and graphics rendering.

Link https://github.com/houdaifa-boukhari/CUB-3D-42-School

Minishell – Unix Shell Implementation

Team Project | Language: C | **Domain:** System Programming, Shell Development, Linux
Designed and implemented a custom Unix shell replicating core Bash functionalities.

- Developed the full execution engine: command parsing, piping (|), redirections (<, >, <<, >>), and environment variable management.
- Integrated signal handling for SIGINT, SIGQUIT, and EOF (Ctrl+C, Ctrl+D, Ctrl+\), maintaining behavior close to Bash.
- Ensured robust memory management and error handling, following strict coding standards and memory leak prevention.

Delivered a lightweight, stable shell built entirely in C with clean process and signal control.

Link https://github.com/houdaifa-boukhari/mini-shell

NetPractice – Network Architecture Simulator

- Configured and troubleshooted 10 progressively complex TCP/IP network diagrams via a custom web interface, mastering addressing, subnetting and router integration.
- Automated submission and peer-evaluation workflows by exporting each level's configuration to Git, ensuring reproducibility and version control.

CTF EXPERIENCE & CYBERSECURITY ENGAGEMENT – 2025

2024 – CURRENT

Active Participant in Capture the Flag (CTF) Competitions – 2024–2025

- **3rd Place – Campus-Wide CTF Competition**
  1337 Coding School, Tétouan, Morocco – November 2024
  Earned 3rd place among 25 teams, showcasing advanced skills in reverse engineering, binary exploitation, and cryptographic analysis.

- **4th Place – ENSA Tanger CTF**
  École Nationale des Sciences Appliquées de Tanger, Morocco – December 2024
  Demonstrated expertise in memory corruption exploits and crypto-based challenges, securing 4th place in a competitive regional event.

- **3rd Place – ENSA Tétouan CTF**
  École Nationale des Sciences Appliquées de Tétouan, Morocco – February 2025
  Achieved 3rd place with a strong performance in complex binary exploitation tasks and reverse engineering problems.

- **7th Place – ENSIAS Rabat CTF**
  École Nationale Supérieure d'Informatique et d'Analyse des Systèmes, Rabat, Morocco – April 2025
  Ranked 7th nationally, gaining deeper experience in advanced reverse engineering and cryptographic challenge solving.